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The Physical Basis and Origin of Hierarchical Control

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My study of hierarchical organization began from the point of view of a physicist interested in the origin of life. I would not choose to study a concept that is so general and so poorly defined, except that no matter how I have looked at the question of the origin of life, I have not been able to evade the problem of the origin of hierarchical control. Hierarchical organization is so universal in the biological world that we usually pass it off as the natural way to achieve simplicity or efficiency in a large collection of interacting elements. If asked what the fundamental reason is for hierarchical organization, I suspect most people would simply say, "How else would you do it?"

This existential response has the right hierarchical spirit, for indeed one central result of hierarchical organization is greater simplicity; and yet any analytical approach to understanding simplicity always turns out to be very complex. We do not really mean just "simplicity" but *functional* simplicity. The elegance of a physical theory or a work of art depends on simplicity, but never on simplicity alone. There must also be a measure of effectiveness. In the same way, the simplification that results from the hierarchical constraints of an organization must be balanced by how well it functions.

What are the central problems about hierarchical systems? First there is the apparent paradox that hierarchical controls both limit freedom and give more freedom at the same time. The constraints of the genetic code on ordinary chemistry make possible the diversity of living forms. At the next level, the additional constraints of genetic repressors make possible the integrated development of functional organs and multicellular individuals. At the highest levels of control we

know that legal constraints are necessary to establish a free society, and constraints of spelling and syntax are prerequisites for free expression of thought.

A second problem about hierarchical constraints is that they always appear arbitrary to a large extent. As far as we can see, the same type of life could exist with a number of different genetic codes—that is, with different assignments of nucleic acid codons to amino acids. Molecules that perform the function of messengers, such as hormones or activator molecules, appear to have only arbitrary relation to what they control. Other hierarchical rules are more obviously conventions. We know we can drive on either the left or right side of the road as long as there is collective agreement, just as we know that we can give the same instructions in many different languages using different alphabets. In other words, hierarchical constraints or rules are embodied in structures that are to some extent “frozen accidents.”

Even in the arts when there are no chemical or cultural constraints, the artist must invent his own. Igor Stravinsky writes in *Poetics of Music*, “The more constraints one imposes, the more one frees one’s self of the chains that shackle the spirit . . .” and he goes on, “. . . the arbitrariness of the constraint serves only to obtain precision of execution.”

In the inanimate world, there are certain types of constraints that produce structures that we recognize as atoms, molecules and crystals, and eventually mountains, planets and stars, but these constraints are not associated with new freedom in the collections, and even though some of the shapes of crystals and mountains appear arbitrary, these shapes could not be associated with any kind of effectiveness of function or precision of execution. This is in sharp contrast to all the hierarchical control levels of living matter, and

for this reason any theory of hierarchical origins must explain the origin of the type of constraints that are both arbitrary and effective in the sense of giving freedom to the collection.

In contrast to the earlier chapters, it is going to be my strategy to approach these problems primarily from the point of view of physics rather than cellular or developmental biology, sociology, or ecology. At first this may appear as an unlikely strategy since hierarchical organization is foreign to most physics, but common to all the biological sciences. On the other hand, if we want to understand origins we must begin at a simple enough level so that hierarchical controls are not already inherent in the behavior of the system. We must ask what the first level is in the physical world where arbitrary constraints can produce a new kind of freedom of behavior.

Structural versus Control Constraints

What I am really asking for is the physical basis for the origin of life, which could be taken as the same problem as the origin of those control constraints that free living matter to evolve along innumerable pathways that non-living matter, following the same detailed laws of motion, cannot follow. In other words, although we recognize structural hierarchies in both living and non-living matter, it is the *control* hierarchy that is the distinguishing characteristic of life. Both structural and control hierarchies are partial orderings, like Simon's Chinese boxes, but the concept of control hierarchy is a narrower partial ordering that implies an ac-

tive authority relation of the upper level over the elements of the lower levels. It is this original functional meaning I want to preserve, but without the original Divine implications.

Before trying to give a physical description of control, let me briefly review what we mean by a structural hierarchy. Simon, in Chapter 1 of this volume, has given an excellent description of the physical basis of structural hierarchies as well as a theory of their origin. His theory depends on the relative assembly times for elements with certain probabilities of association and dissociation. This time is drastically shortened if there exist stable substructures. Thus we find development of multicellular organisms based on single cells with high autonomous stability. In the same way, crystals are formed from stable atoms, and words are formed from stable alphabets.

The atom, the molecule, the crystal, and the solid can be distinguished as levels by the criterion of *number*; that is, each level is made up of a large collection of the units of the lower level. However, there is also a physical hierarchy of *forces* underlying these levels, the strongest forces being responsible for the smallest or lowest level structures. The strongest force holds together the nuclei of the atoms, and the weakest force, gravity, holds together the largest bodies of matter. There is also a hierarchy of dynamical *time* scales, which may be associated with the levels of forces, the shortest time with the strongest force and smallest structures, and the longest time with the weakest force and largest structures.

As a result of these graded levels of numbers, forces, and time scales, we can often write dynamical equations for only one level at a time using the approximations that one particle

is typical or representative of the collection, that the fast motions one level down are averaged out, and that the slow motions one level up are constant. It is this type of approximate treatment of the dynamics of many hierarchical structures which Simon calls "near-decomposability." The simplicity and solvability of most physical equations depend on making these approximations. In structural hierarchies this interface can usually be ignored, except for the analysis of the errors of the single-level approximation.

Hierarchical control systems are not this simple. In a control hierarchy the upper level exerts a specific, dynamic constraint on the details of the motion at lower level, so that the fast dynamics of the lower level cannot simply be averaged out. The collection of subunits that forms the upper level in a structural hierarchy now also acts as a constraint on the motions of selected individual subunits. This amounts to a feedback path between levels. Therefore the physical behavior of a control hierarchy must take into account at least two levels at a time, and what is worse, the one-particle approximation fails because the constrained subunits are atypical.

For example, the development of multicellular organisms, which is discussed by Grobstein and Bonner in chapters 2 and 3, shows that the cells do not simply aggregate to form the individual, as atoms aggregate to form crystals. There are chemical messages from the collections of cells that constrain the detailed genetic expression of individual cells that make up the collection. Although each cell began as an autonomous, "typical" unit with its own rules of replication and growth, in the collection each cell finds additional selective rules imposed on it by the collection, which causes its differentiation. Of course, this is also the general nature of social control hierarchies. As isolated individuals we behave in

certain patterns, but when we live in a group we find that additional constraints are imposed on us as individuals by some "authority." It may appear that this constraining authority is just one ordinary individual of the group to whom we give a title, such as admiral, president, or policeman, but tracing the origin of this authority reveals that these are more accurately said to be group constraints that are executed by an individual holding an "office" established by a collective hierarchical organization.

In a similar way, developmental controls in cells may be executed by "ordinary" molecules to which we give titles, such as activator, repressor, or hormone, but the control value of these molecules is not an inherent chemical property; it is a complex relation established by a collective hierarchical organization requiring the whole organism. At the lower level of the gene, the authority relation of the hierarchy is often popularly expressed by referring to DNA as the "master molecule" of life, but here again we must emphasize that there is no intrinsic chemical property of DNA that allows it to hold this office. It is the integrated collection of "ordinary" molecules we call the cell that endows DNA with this authority. We should not expect that more detailed study of DNA, enzymes, and hormones would reveal other than ordinary molecules any more than we would expect that a detailed study of presidents would reveal other than ordinary men. The interesting problem of the origin of hierarchical control is to explain how such ordinary molecules and men can evolve such extraordinary authority as members of a collection. Or to put the problem in other words, how do structures that have only common physical properties as individuals achieve special functions in a collection? This statement of the problem shifts the emphasis from one level

or another to the hierarchical *interface* between levels.

Rosen (1969) has expressed the problem in a very similar way by defining a hierarchical organization as a system that has more than one simultaneous activity (i.e., structure and function) such that alternative modes of description are an absolute necessity. As I shall show, this is also a characteristic of the physical concept of constraint.

The Structure-Function Problem

For many reasons, I wish I could evade this classical problem of the relation of physical structure to biological function. One reason is that it has generated so many polemics associated with reductionist, vitalist, and teleological arguments. A second reason is that it is currently out of favor as a biological problem. There is some reason for this. Almost all of the discoveries that make up what is called the molecular biological revolution of the last 20 years have been generated by a commitment to a strategy that says that to really understand biological function one must know the molecular details of structure. It was the persistent search for the underlying molecular structures in biochemistry and genetics that has produced our present descriptions of how cells function. Therefore, most biologists today hold strongly to the strategy of looking at the molecular structures for the answers to the question of "how it works."

Nevertheless, it is surprising and discouraging to find so many biologists who, finding this strategy productive, mistake it for a theory of life. Some biology departments have even gone so far as to exclude study of theories of life, as if

the detailed facts of molecular biology had somehow demonstrated that theory is not relevant for biology. I was once asked by a leading molecular biologist, quite seriously, "If we can find all the facts, why do we need a theory?" This attitude is especially inappropriate now that molecular biologists are moving on to developmental biology and neurobiology where the integrated function is removed from the detailed structure by even more hierarchical control interfaces. One could not imagine a mathematician trying to understand the nature of computation in terms of how real computer components are designed and wired together. In fact, deep understanding of the nature of computation has come only from theories of computation, which are largely free of the details of real machines. As of now there is no corresponding theory through which we can understand the nature of life, but I shall argue that the study of hierarchical control will form the basis for such a theory.

Let me make it clear that I am not minimizing the importance of collecting the essential detailed biochemical facts of life any more than I would minimize the importance of knowing how to design real switching and memory devices that work with high speed and reliability in a computer. What I wish to emphasize is that the structure-function problem is still very much with us in biology, in spite of our new knowledge of molecular details. I shall try to show that this structure-function duality arises inevitably at a hierarchical control interface. We cannot understand this interface by looking only at structural details or at the functional organization. The problem is precisely at the interface between the detail of structure and the abstraction of function. In fact, what I shall conclude is that function or control can only arise through some selective loss of detail. The problem,

which is especially acute for the physicist who believes that nature takes care of all her details, is to explain how a natural "selective loss of detail" can lead to hierarchical control instead of the usual loss of order in the system.

Remember, we are looking for a physical reason why an ordinary molecule can become the controlling factor in forming a chemical bond or in the expression of a whole developmental program. A control molecule is not a typical molecule even though it has a normal structure and follows normal laws. In the collection where it exerts some control it is not just a physical structure—it functions as a *message*, and therefore the significance of this message does not derive from its detailed structure but from the set of hierarchical constraints which we may compare with the integrated rules of a language. These rules do not lie in the structure of any element. We are asking for the physical basis of the hierarchical rules of the collection that turn these ordinary molecules into special messages. Or, to put it in the older terminology, we are still looking for the physical basis of the origin of function in collections of molecules. I prefer, however, to use the concept of hierarchical control rather than hierarchical function, since control implies the constraint or regulation of a total system, whereas function often applies to a specific process that is only a small part of the whole organism. I shall continue, then, by trying to say more clearly what control implies in elementary physical language.

What is a Control Device?

How are we to recognize the simplest examples of hierarchical control? How complex must a physical system be in order to exhibit control? Does control appear gradually, or is there a discrete threshold? To answer these questions we must specify more precisely what we mean by "control" in the context of biological origins and evolution. Let me begin to do this with some simple examples. First, consider an idealized process of crystal growth. Suppose you begin with a glass of water with common salt in solution. The sodium and chloride ions are free to move about in three dimensions. We say "free" only in the sense that they each follow the laws of motion of non-interacting particles with only occasional collisions with other molecules or ions. More precisely, we say that most of the time each particle has three translational degrees of freedom. Now suppose that after some time a collection of ions has formed a substantial crystal. This structure can now act as a constraint for some of the ions that land on one of its surfaces. These ions now have fewer degrees of freedom, like a ball constrained to the floor instead of moving through space. It is a collective constraint on individual elements that make up the collection. Do we mean, now, to speak of the crystal as a natural hierarchical control device?

I think not. My purpose in giving this example is to suggest that the concept of control involves a more active dynamical role than simply limiting the available space in which matter can move. The evolution of such time-independent constraints—that is, the addition of more such constraints in the course of time—can lead only to fixed structures, the crystalline solid being a good example. A

more realistic variation of this example is the screw-dislocation crystal growth, which is a common process by which crystals actually grow. Instead of each ion binding only at the points of a perfect lattice, there are imperfections in growth which produce a new kind of constraint known as the screw-dislocation. This constraint has two properties: (1) It speeds up the binding of ions by an enormous factor, and (2) it preserves its screw structure as the crystal grows. This persistent speeding up of the rate of growth as the result of the collective structure of the screw dislocation is closer to an active control process.

However, the end result of adding such constraints is still a relatively rigid and permanent structure which clearly does not have the potential for evolution that we associate with living matter. There are many examples of the growth of non-living structures that have this final rigidity. The general difficulty is that we need to find how to add constraints without using up all the degrees of freedom. In other words, what we need for a useful control system is a set of constraints that holds between certain degrees of freedom, but that does not lead to completely rigid bodies. Of course, lack of rigidity is not sufficient; for example, a balloon constrains the gas inside it without freezing up into a rigid body, but a balloon does not have enough constraint on the motions of the gas molecules to be called a control device. A physicist would call the balloon a boundary condition.

What we need to find, then, is a clearer description of the degree of constraint that gives rise to a control hierarchy. We can state two conditions that must be satisfied. First, an effective control event cannot be simply a passive, spatial constraint, but must actively change the *rate* of one particular event, reaction, or trajectory relative to the unconstrained

rates. This condition is fulfilled by most devices that we normally associate with existing control systems—for example, switches and catalysts. Second, the operation of the constraint must be *repeatable* without leading to the freezing up of the system. Another way to say this is that control constraints must limit the trajectories of the system in a regular way without a corresponding freezing out of its configurational degrees of freedom. In physical language this condition is satisfied by a *non-holonomic* or non-integrable constraint (e.g., Whittaker, 1936). Every interesting man-made control device must be represented as a non-holonomic constraint—switches, ratchets, escapements, and gear shifts being common examples. But at this point we must be very careful not to evade the problem by formalizing it. “Non-holonomic constraint” is acceptable jargon in physics, just as “derepressor” is acceptable jargon in biology. We may state clearly a mathematical function representing a constraint just as we may state clearly the function of a derepressor molecule. But such functions are not derived from what we call the fundamental laws; they are only added on as needed to describe the behavior of integrated systems. In physics the notion of constraint is not considered a fundamental property. It is not useful at the atomic or astronomical scale where the forces between “free” particles are sufficient to describe the motions. So even though we have another word to describe control, we have no idea of how control constraints actually originate. What we need to do is look more closely at the physical basis of natural constraints—how they can arise spontaneously and how they can be classified into structure-producing and control-producing relations.

What Is a Constraint?

The common language concept of a constraint is a forcible limitation of freedom. This general idea often applies also in mechanics, but as we emphasized in the beginning, control constraints must also create freedom in some sense. Also we must distinguish the forces that enter in the dynamical laws of the system and the forces of constraint. For example, there is the force of gravity and electric fields that enter into the equations of motion and determine how the system will move in the course of time. These fundamental forces do indeed “limit the freedom” of the particles, but the fact is that they *leave the particles no freedom at all. Or to put it more precisely*, the forces that enter the equations of motion determine the change in time of the state of the system as closely as determinism is allowed by physical theory. The whole concept of physical theory is based on the belief that the motions or states of matter are neither free nor chaotic, but governed by universal laws. So what meaning is there in our concept of additional “constraints” or additional “forceful limitations” on matter when theory says that no additional constraints are possible on the microscopic motions?

The answer is that the physicist’s idea of constraint is not a microscopic concept. The forces of constraint to a physicist are unavoidably associated with a new hierarchical level of *description*. Whenever a physicist adds an equation of constraint to the equations of motion, he is really writing in two languages at the same time. The equation of motion language relates the detailed trajectory or state of the system to dynamical time, whereas the constraint language is not about the same type of system at all, but another situation in which

dynamical detail has been purposely ignored, and in which the equation of motion language would be useless. In other words, forces of constraint are not the detailed forces of individual particles, but forces from collections of particles or in some cases from single units averaged over time. In any case, some form of statistical averaging process has replaced the microscopic details. In physics, then, in order to describe a constraint, one must relinquish dynamical description of detail. A constraint requires an *alternative description*.

Now I do not mean to sound as if this is all clearly understood. On the contrary, even though physicists manage quite well to obtain answers for problems that involve dynamics of single particles constrained by statistical averages of collections of particles, it is fair to say that these two alternative languages, dynamics and statistics, have never been combined in an elegant way, although many profound attempts have been made to do so.¹ Furthermore, the problem has proven exceedingly obscure at the most fundamental level—namely, the interface between quantum dynamics and measurement statistics. This is known as the problem of quantum measurement, and although it has been discussed by the most competent physicists since quantum mechanics was discovered, it is still in an unsatisfactory state. What is agreed, however, is that measurement requires an *alternative description*, which is not derivable from quantum dynamical equa-

1. How well the dynamical and statistical descriptions have been combined is, of course, a matter of opinion. The basic problem is that the dynamical equations of matter are strictly reversible in time, whereas collections of matter approaching equilibrium are irreversible. The resolutions of this problem have been central to the development of statistical mechanics and have produced many profound arguments. For our purposes we need not judge the quality of these arguments, but only note that the resolutions always involve *alternative descriptions* of the same physical situation. (See e.g., Uhlenbeck and Ford, 1963)

tions of motion.² Bearing in mind that even the clearest physical picture of a constraint involves a hierarchical interface which needs more careful analysis, we may distinguish some of the more common types of structures that we observe to originate spontaneously. We are still looking for conditions that would favor control constraints, but we must begin with the better understood structural constraints.

The chemical bond is undoubtedly the most fundamental structure in the energy range where we recognize life processes. But is it useful to consider this structure as a constraint in the language of control systems? One could argue that since the structures of atoms and molecules are stationary states, they are in a definite state of precise energy, and therefore time is excluded by the uncertainty principle. Thus, we might say that to describe this as a constraint we relinquish the detailed time description of the motion. However, in this case, the structure is really a solution of the equations of motion, and the fact that the energy and time are not simultaneously measurable is not the result of collective forces or an averaging process, but an essential condition of the fundamental dynamical language of quantum mechanics. In other words, the laws determine the structures—there is

2. The quantum measurement problem is closely related to the statistical irreversibility problem, and it too has a long history of profound arguments central to the interpretation of quantum theory. The basic difficulty here is that a physical event, such as a collision of two particles, is a reversible process, whereas the record of this event, which we call a measurement, is irreversible (the record cannot precede the event). Yet, if we look at the recording device in detail, it should then be reducible to reversible collisions between collections of particles. Again for our discussion here it is not necessary to judge the many attempts to resolve this difficulty, since as a practical matter they all involve *alternative descriptions* for the event and the record of the event. (See, e.g., von Neumann, J., 1955, for a mathematical treatment, or Wigner, E., 1963, for a non-mathematical review of the problem. For a discussion of quantum measurement and biology, see Pattee, H. H., 1970).

no alternative. I would, therefore, prefer not to call a single chemical bond a constraint, although it is certainly a structure.

But clearly all constraints depend on chemical bonds. A billiard table is held together by such bonds, and it is called a constraint because the motions of all the atoms in the table can be averaged out to form relatively simple limits on the motion of the billiard balls. So the question arises: How many chemical bonds do we need before we can speak of a constraint? Of course there is no exact number, such as 10 or 10,000. I believe there is only a very pragmatic answer that one can give: I would say that *a dynamical collection is described as a constraint when there exist equations or rules in a simpler form that direct or control the motions of selected particles*. Of course the dynamical equations must still tell us *in principle* how the whole system will evolve in time, without involving the concept of constraint.

In any case, our conclusion is that an equation of constraint in physics is an *alternative description* of the microscopically complex and deterministic motions that gains in simplicity or utility by selectively *ignoring certain dynamical details*. In effect, the physicist has *classified or coded* the microscopic degrees of freedom into a smaller number of new variables. How he performs this classification is not predictable in any dynamical sense, since this depends on the choice of problem he wishes to solve and the degree of approximation he is willing to accept. At the same time, it is also true, as we have mentioned, that the structures of nature fall into “nearly-decomposable” levels that differ in the magnitudes of their forces, sizes, numbers, or time scales, so that the physicist’s classifications for a given problem often appear obvious and “natural.” But this is a long way from a

truly spontaneous classification, which could occur without a highly evolved living system. This kind of classification or selective neglect of detail still requires an intelligent physicist with a problem to solve.

However, I wish to generalize this concept of constraint so that it would apply even before physicists existed. Let me say that a hierarchical constraint is established by a particular kind of new rule that represents not merely a structure but a *classification* of microscopic degrees of freedom of the lower level it controls. The classification may take many forms, which an intelligent observer might define as averages of microscopic variables, or as selection of a few sensitive degrees of freedom, or as a redefinition of the system. But in some sense, the appearance of a natural constraint implies an internal classification process that is selected on the basis of simplicity, utility, or *function* of this alternative description. Now we are ready to ask what types of constraints will lead to hierarchical control. Can we state some principles of hierarchical control?

The Principle of Classification of Details

By looking at the idea of control and constraint in physical language we have concluded that a hierarchical control level is established by a particular kind of constraint that represents not simply a structure but a classification of the details of the lower level. Before asking what a natural classification could be and how it may arise, I would like to show how this principle of hierarchical control applies generally. Is it reasonable to claim that the selective neglect of certain details

is a universal property of hierarchical control systems? Let us consider some examples.

What about the hierarchical control of the developmental processes? Here the lowest level is the collection of self-replicating cells. Bonner found that to represent the developmental process by a program it was necessary to use the concept of the *developmental test*. According to this concept, the developing organism performs tests of the environment or surrounding cells, and the outcome of the tests is to turn off or on the genes appropriate for the developmental response. Now clearly such "tests" must classify interactions. First, there must be a selection of what is tested. For example, such tests would not measure the positions of all the amino acids in the environment—that would hardly be significant for the cell even if it were practical. Second, there must be a selection of what range of results of a test will trigger a control response. Thus, out of the innumerable detailed physical interactions of the cells and their surroundings, there is a classification into significant and insignificant interactions, which I would say amounts to selective neglect of details in favor of only a very limited number of crucial conditions. Goodwin and Cohen (1969) say it is as though every cell need only read a clock and a map, but it is the classification scheme that creates the right "clock" and right "map."

Grobstein's description of the process of folding in the protein molecule provides perhaps the most fundamental example of primitive hierarchical organization. This folding of the linear chain of amino acid residues creates a new property of the chain—the enzymatic function. At the lower level of chain construction, each amino acid is recognized by its transfer enzyme and RNA and placed in sequence under

the control of the messenger RNA. Here there is no doubt that certain detailed structures identifying each amino acid must be crucial for accurate construction, whereas on the next level up, when we can use an alternative description of the amino acid collection appropriate to its *function* as an enzyme, such detail is no longer necessary. In fact, there is good experimental evidence that there are equivalence classes of amino acids in the chain which allow the same enzymatic function (e.g., see Nolan and Margoliash, 1969).

At the other extreme of complexity, in human organizations and political hierarchies, there are also many examples of the selective ignoring of details. Indeed, at no other level do the rules of classification and selection of what we ignore appear so crucial to the stability and quality of the collective society. It is significant that these human rules about what we ignore are often promulgated as Divine moral principles or at least self-evident conditions for freedom and justice. The principle of "equality under the law" is at the foundation of most legal forms of hierarchical control, and this is little more than an abbreviated statement that legal controls should apply, not to individuals, but to equivalence classes of individuals, where most details must be ignored. An extreme example is the law that says that anyone driving through a red traffic light will be fined. Think of the innumerable details that this control implicitly ignores! Some controls also state explicitly what details we shall ignore, such as laws against racial or religious discrimination.

However, social hierarchies show us that legal controls may either ignore too many details or too few. In the case of traffic control, it is well known that a simple, fixed "constraint" of a four-way stop sign is not as effective in moving traffic as the traffic light that operates according to certain

inputs from sensors in the road. In this case, the sensitivity to more details of the motion of vehicles results in increased traffic flow. But it is also clear that a traffic light which took account of too much detail, such as the number of people in the car, could not effectively increase traffic flow. In the same sense, a law that simply requires executing all individuals who have killed another person ignores too many details to be acceptable to society. A few more details must be considered such as the age and mental health of the individual as well as the state of society (i.e., whether the society is "at war"). But again, too much detail would make the control ineffective. The optimum amount of control detail must depend on the desired function. Punishment for murder does not consider the type of weapon used, whereas punishment for robbery does.

The Principle of Optimum Loss of Detail

We have looked at hierarchical control now at several levels, from crystal growth up through cells to societies. At the extremely simple level of atoms and crystals we argued that the concept of control implied a selective rate control process performed by a collection on individual elements. The control event is repeatable for different elements without freezing up the collection. This type of rate control is accomplished by flexible (non-holonomic) constraints that can be neither too tight nor too loose. If they are too tight, we see more or less rigid bodies with no "function," whereas if they are too loose we see only "boundary conditions" that have no definite, repeatable effect on the elements of the collection.

At the other extreme level of complexity, in human social controls, we also observe that too much constraint on too many details leads to what we might call bureaucratic “freezing up” of the system, whereas too few constraints leads to ineffective function, and ultimately to anarchy. It is easy to imagine how similar difficulties would arise at the enzymatic, cellular, and developmental levels of control if the constraints, or classifications they represent, are either too detailed or too general. Function cannot arise in a strictly deterministic system or in a completely random system.

This situation strongly suggests a second hierarchical control principle. We shall see later that this principle is especially important for understanding hierarchical origins. The principle states that hierarchical control appears in collections of elements within which there is some optimum loss of the effects of detail. Many hierarchical *structures* will arise from the detailed dynamics of the elements, as in the formation of chemical bonds, but the optimum degree of constraint for hierarchical *control* is not determined by the detailed dynamics of the elements. The dynamics of control is determined by how these details are ignored. In other words, *hierarchical controls arise from a degree of internal constraint that forces the elements into a collective, simplified behavior that is independent of selected details of the dynamical behavior of its elements*. It is the combination of the *independence* of the constraints on the microscopic dynamics along with the *simplification* of the collective dynamics which creates what we recognize as integrated behavior or function. This optimum level of constraint cannot be specified precisely—that is one of our problems—but it falls between the extremes of strict determinism on the one hand and the chaos of “equiprobability of accessible states” on the other. It is sig-

nificant that the concept of "freedom" has meaning only in behavior that falls between these same extremes, and this helps us understand how freedom can sometimes be generated by adding constraints and other times by removing constraints.

It follows that the origin of an *autonomous* hierarchical control requires that the independence from detail arises spontaneously from this detail. That is, the microscopic dynamics at the lower level must lead to structures with macroscopic dynamical behavior that is independent of microscopic initial conditions, at least over some finite range. What kind of system has this property?

The Principle of Statistical Closure

When we think of simple dynamical trajectories, such as a baseball thrown to first base or a free satellite coasting toward the moon, we know that success or failure depends crucially on the precision of the initial conditions—that is, the initial position and velocity of the projectile and the position and velocity of its target. This is not the kind of dynamical behavior that leads spontaneously to hierarchical control. At the other extreme, we are familiar with chemical reaction systems in which, no matter what the detailed initial conditions of each reactant molecule, the whole system reaches equilibrium in a very short time. This is not a dynamical but a statistical process, and the theory involved is sometimes characterized as a description of just those aspects of the system that do not depend on initial conditions. Again, we know that this pure statistical behavior does

not lead to spontaneous hierarchical control. In dynamical systems, any error in initial conditions eventually leads to chaos, while in statistical systems only chaos in initial conditions leads to simple behavior. What we need for hierarchical control is something in between these extremes, or more precisely, a combination of both.³

We have seen that hierarchical controls arise from a collection of elements, but act on individuals of the collection. The spontaneous appearance of collective constraints must therefore be a statistical property of the collection, but its control behavior must act on the dynamics of individual elements. The collective constraint cannot be a dynamical process, since its effect would therefore depend strongly on the initial conditions of its components, which would in effect make it unrecognizable as a constraint. In other words, as we argued earlier, a constraint has no meaning if it cannot be expressed as a simpler alternative description affecting the detailed dynamics. Such an alternative description must have a statistical character, as we have seen, but at the same time this collective behavior of the constraint must harness the detailed dynamical motion of the elements making up the collection.

This harnessing of the lower level by the collective upper level is the essence of hierarchical control. It establishes the

3. Other types of statistical behavior in between the dynamical and equilibrium statistical systems show a certain degree of self-organization. In strongly non-equilibrium systems, when there is a steady energy flow or dissipation, there is the possibility of oscillations, cycles, and spatial patterns of flow or chemical reaction which could be called hierarchical control (See e.g., Burgers, 1963; Morowitz, 1968; Prigogine and Nicolis, 1971). However, this type of control constrains only statistical variables of the system and does not directly limit the dynamical variables of individual elements of the collection, as in the residue-by-residue ordering in proteins and the switching on or off of single genes in individual cells. (See Pattee, 1970, for further discussion of this difference.)

“authority” relation between levels within a closed system. It is common to have structural hierarchies beginning with elements and passing to collections of elements, on up through collections of collections, and so on, but such structural hierarchies are open-ended; they do not close back on their lower levels to establish the control-authority relations. (Of course we may also impose controls from outside, but then we are not talking about closed systems or spontaneous origins.) This necessity for the collective, statistical constraint to close back on the internal, detailed dynamics of the elements of the collection I shall call the *Principle of Statistical Closure*.

Now at this point, one might ask if “statistical closure” implies any more than our original definition of a hierarchical control system—namely, a set of collective constraints that in some way limits the detailed behavior of the elements that make up the collection. What I want to do by using this slightly obscure phrase instead of the longer definition is to emphasize what I interpret as the most fundamental aspect of hierarchical control. It is in the nature of a “juxtaposition of disparate categories,” to use Koestler’s condition for a creative act. Closure is a concept defined in abstract mathematics that is a kind of constraint on the type of rules by which elements of a set can be combined so that no new elements are created. By the use of the closure concept here, I mean a collection of element types that may combine or interact with each other individually in many ways, but that nevertheless persist as the *same* collection when looked at in detail over a long period of time. The mathematical concept of closure is a discrete concept that has nothing to do with the *rate* of combination of the elements in the set. That is, the mathematical idea of closure is entirely independent of

time. Similarly, mathematical closure does not involve the possibility of error in local combinations or the number of trials over an extended time interval.

What I mean by *statistical* closure, on the other hand, is a collection of elements that is established and that persists largely because of the *rates* of their combination. This in turn implies a *population dynamics* for the elements and therefore a real-time dependence. Furthermore, the rates of specific combinations of elements must be controlled by collections of the elements in the closed set.

Examples of Statistical Closure

Since this characterization of hierarchical control is somewhat abstract, I shall give some examples from different levels of biological organization to support the principle that collective statistical constraints do, in fact, establish a closure property on the elements of the hierarchical collection. Perhaps the best known hierarchical control process in biology is at the evolutionary level, where the elements are the individual organisms and the closed collection is the "breeding population." It is a well-known principle of evolution that natural selection does not operate deterministically on individuals, but statistically on the breeding population. The effect of natural selection, however, must be recorded in the hereditary memory of the individual organisms. Therefore, selection is a collective constraint that limits the detailed structure of the individual elements of the collection and establishes a statistically closed set, which we call the breeding population.

At the next lower level we know that the individual is made up of many cells. Here it is not so easy to see what aspects of the development of the multicellular organism are statistical, or even that any statistics are involved at all. Many developmental biologists have adopted the language of computer science to describe the developmental process, as does Bonner in Chapter 3 of this volume, and this might suggest that only deterministic rules are needed to establish the closed collection of cells that forms the individual. Now it may be possible to imagine a strictly discrete, deterministic automaton model of biological development, in which each cell locates its position in the collection of cells by a mechanical test on the fit or state of nearest-neighbor cells, and then turns off or on its appropriate genes according to its developmental program. No statistics would appear to be involved, at least in the formal description of such a model, since the fit or state of other cells might be formally treated as a discrete, mechanical property.

But apparently nature does not do it that way. The individual cell switches off or on or regulates much of its growth according to *concentrations* or *gradients* of concentrations of "message molecules," and this is fundamentally a *statistical* process. Wolpert's (1969) "positional information" is a field concept that requires number averages. Even if some genetic controls are effected by temporal patterns of activity, as in Goodwin's and Cohens's (1969) phase-shift model of development, the concept of *phase* must involve a number or time averaging, which again is a statistical rather than a "mechanical" process. Since we lack so much detailed knowledge of developmental programs, we cannot go much further with this example. However, from what we know already, the developmental process appears to require statistical con-

straints formed by a *collection* of cells which establishes a detailed control on the genetic expression of *individual* cells that make up the collection.

Looking now at the cell itself, we reach the most fundamental hierarchical control level, where a small number of monomer types combine to eventually produce a replicating collection. The closure property is obvious, but where are the essential statistical constraints in the cell? Here again, as in the models of development, we can construct self-replicating automata that do not appear to have any statistical process or behavior whatsoever. Even descriptions of self-replication in the language of molecular biology contain no words or concepts that are as obviously statistical as "concentration" or "gradient." Many biology texts presently describe the replication process as entirely deterministic, discrete, mechanical events at the molecular level. Accordingly, one could "design" a cell, using only *one* molecule of each essential nucleic acid and enzyme that would reproduce itself by the same general replication, transcription, translation, and synthesis scheme that is used for real cells, although a certain amount of inefficiency might result. So how can statistical processes arise when only one of each element is necessary?

The answer, I believe, is in the statistical nature of the remarkable molecular devices that make this "scheme" work—the enzymes. Without enzymes there would be no DNA replication, no transcription to RNA, no coding, and no synthesis. The enzymes control the *rate* of selected reactions, and rate control is not a dynamical, but a statistical concept. The reason we do not so easily see the statistical behavior is that selective rate control in enzymes is accomplished by time-averaging rather than number-averaging, which we use to define concentrations. The enzyme (or a more primitive

macromolecule with selective catalytic function) is about the simplest, or at least the smallest, structure that fulfills the conditions for a constraint given in a previous section—namely, a complex dynamical system whose functional behavior requires an alternative description of a simpler form. We said this behavior in a hierarchical constraint results in the control of the detailed motions of selected degrees of freedom. This is precisely the nature of enzymes. At the functional level, we speak of the enzyme “recognizing” its substrate, which means that its many collisions with other molecules have no regular effect, and even many degrees of freedom of the substrate are ignored in the binding and catalytic steps. However, at the functional level the operation of the enzyme is specific and repeatable—the same enzyme molecule producing the same catalytic reaction over and over.

Therefore, although any microscopic dynamical description of the enzyme and substrate must follow the reversible laws of motion, at the functional level of description we say the enzyme recognizes the substrate, catalyzes a bond, and resets itself; and all three of these operations are inherently irreversible and hence statistical in some sense. Here both our empirical knowledge of enzymes as well as our ability to construct quantitative models of selective catalysts is weak, and much more thought and experiment will be needed to understand this fundamental type of constraint.

Even though enzyme function is a statistical constraint, it is not by itself an example of statistical closure, since a single enzyme could not selectively constrain all the elements—the amino acids—of which it is made up. In fact, the smallest collection we can now observe which has the statistical closure property is the cell, for in the cell, each monomer of the

nucleic acids and proteins is specifically constrained in its reactions by one or more copolymers made from this same collection of monomers. But the cell is too complex to have come into existence spontaneously. *The general hierarchical origin problem is to find the simplest or most probable natural collection of elements which exhibits statistical closure.* At the origin of life, the elements were very likely some set of amino acids, nucleotides, or both. At higher levels, of course, the elements are increasingly complex, such as cells or individual organisms. But at each new level there must have been a minimum set of conditions to generate a new statistical closure.

The Origin Problem

No matter how closely we describe existing hierarchical organizations or how much we appreciate their efficacy, their origins present a new type of problem. First of all, it is difficult to find examples of incipient hierarchical control or observe the spontaneous growth of hierarchical control. Human social hierarchies can be studied historically, but the forces here are so complex and always involve so many intuitive or rational decisions of the human intellect, that any relation with primitive hierarchical origins is thoroughly obscured.

In the field of evolutionary biology we can find many experts who take a rather relaxed and optimistic view of the potential for self-replicating hereditary units to discover new and more effective hierarchical organization through blind mutation and natural selection. On the other hand, there

have always been critics who do not find the theory of natural selection a satisfying or sufficient explanation for the *major* evolutionary innovations that have occurred, in particular the origin of new hierarchical control levels. Recently, a new wave of skepticism has appeared among some mathematicians who have tried to imitate or model evolutionary processes on large computers. These attempts include a great variety of problems and programs, which have names such as problem-solving, learning, self-organizing, adaptive, or evolutionary simulations.⁴ Even though many of these programs have been extremely sophisticated and have performed well on certain types of life-like problems, their evolutionary potential has been non-existent or disappointing. Of course, these programs have built-in hierarchical structures, but in no case is it obvious that a new level of hierarchical control could ever evolve spontaneously through a blind search and selection process.

The many detailed discoveries of molecular biology have not been much help in clarifying origin problems either. In fact, in one sense, knowing the details has only intensified the problem. Following the ideas of Oparin and Haldane and the successful experiments on the spontaneous synthesis of amino acids and other essential pre-cellular molecules, the origin of life appeared at first as a natural and very likely

4. Most biological simulations on computers are aimed at understanding or improving our intellectual abilities; e.g., Feigenbaum, E.A. and Feldman, J., 1963. Several symposia on self-organizing systems, supported by the Office of Naval Research, have covered a wide area of these topics; e.g., Yovits, M.C. and Cameron, S., 1960; von Foerster, H. and Zopf, G.W. Jr., 1962; Yovits M.C., Jacobi, G.T., and Goldstein, G.D., 1962. Examples of more recent computer studies of evolution are Bremermann, H.J., 1967; Reed, J., Toombs, R., and Barricelli, N.A. 1967; Conrad, M. and Pattee, H.H., 1970. For a discussion of the theory of evolution between skeptical mathematicians and evolutionary biologists, see Moorhead, P.S. and Kaplan, M.M., 1967.

event. However, now that we know the intricacies of the genetic code and its associated structures, we are again faced with what appears to be an extremely fortuitous hierarchical organization with no obvious process for its spontaneous origin.

Two rather undeveloped theories have been proposed for the universality of the genetic code, and are therefore closely related to its origin. One, called the stereochemical theory, assumes a direct fit between the structure of the three nucleotides forming the codon and the structure of the corresponding amino acid. The other, called the frozen accident theory, assumes that any codon could have been adapted to any amino acid, but that the particular choice was accidental and remains fixed because any change would have been lethal (e.g., Woese, 1967; Crick, 1968; Orgel, 1968).

To my way of thinking, these two theories evade the central problem of the nature of the statistical closure property, which establishes the *coherence* of the code and hence its *descriptive* generality. In the first place, all forms of codes, rules, or descriptions, even the most abstract and symbolic, must have a definite structural basis just as *functions* of all types must have a structural basis. For example, no matter how many adaptor molecules were required between the codon and its amino acid, each adaptor would in some sense possess a structural fit with its neighbor. If this structural fit were only the result of an inherent chemistry of the copolymers involved, then it is by no means clear why any set of copolymers should recognize precisely the set of monomers of which the copolymers are composed. In other words, a stereochemical theory implies that a *structural* closure property is inherent in the physical properties of matter itself. The stereochemical theory is not consistent, then, with the appar-

ent arbitrariness of other types of hierarchical constraints or *descriptions*, or with the idea of *statistical* closure, which depends on the stability or persistence of time-dependent, rate-control processes in populations of elements.

The frozen accident theory, on the other hand, does not conflict with the idea of arbitrariness in the coding rules, but it does not help us understand the statistical closure property either. It is easy to say that a code will be "frozen" when any change is lethal, but the problem arises in the preliminary stages before the coding constraints appear frozen. The establishment of statistical closure may in fact result in an apparent freezing of structures, although there must be earlier levels of freezing before the time the present complex code was established. What we still need to understand are the minimum conditions for the spontaneous appearance of statistical closure in a collection.

It could be that the study of biological organizations in more and more structural detail at all hierarchical levels will eventually suggest the secret of their origins. Certainly such studies are necessary, but I doubt that they will ever be sufficient for a full appreciation of hierarchical origins. The basic reason for my doubt is that detailed structure alone is never a unique basis for any function or description. As we have argued earlier, the concept of hierarchical constraint or function requires an alternative description or a classification of the dynamical details; and the effect or message value of any control molecule is not an inherent property of its structure, but a collective property of a coherent system of molecules. This coherent system allows what we call the alternative *description*, which constrains selected details at the lower structural level so as to establish *function* at the collective upper level. There is, of course, a structural basis for any

description or classification rule as well as any functional behavior. When we say that the DNA molecule is a description of the cell or the organism, we imply that there are enzymes, RNAs, and ribosomes to read this description. But it is not the structural details of these molecules that is essential; it is the properties of the entire coordinated set of rate-controlling constraints that make sense of each detail. It is because of the coherence of the set that we can speak of a description, and it is because the meaning of the description is in collective function and not in individual elements that their individual structures remain to some degree arbitrary.

Summary and Conclusions

The physical basis of hierarchical control does not lie at any one level of physical theory. This means that we cannot understand the nature of biological hierarchies simply by a finer look at molecular structure, nor by solution of detailed equations of motion, nor by application of non-equilibrium statistical thermodynamics. While each of these physical disciplines is useful for describing a particular level of biological organization, hierarchical control operates between levels and is therefore a problem of the nature of the *interface* between levels.

We have characterized this interface by three concepts we call Principles of 1) Classification of Detail, 2) Optimum Constraint, and 3) Statistical Closure. We assume that at the lower level of any hierarchical organization there is a microscopic dynamics of the elements, described by laws of motion, and deterministic in character. In order to speak of

imposing additional "controls" on a system which is already entirely determined in its detailed description, we must relinquish some of this detail and use an alternative description. This amounts to a classification of the microscopic degrees of freedom—our first principle. This classification must be embodied in or executed by what we call a constraint, in particular a flexible (non-holonomic) constraint, which does not simply freeze-out degrees of freedom, but imposes new functional relations between them. The enzyme molecule is our most elementary example of such a flexible constraint that classifies its complex, detailed collisions according to highly simplified functional rules.

We emphasize that the physical description of this type of constraint is necessarily of a *statistical* nature, even though the functional description may appear as a very simple rule. It is this interface between the deterministic laws of the elements and the statistical constraints of their collections for which we still do not have an adequate theory or formalism, and as one might expect, all higher levels of hierarchical control appear to evolve from this primeval level.

Nevertheless, we can recognize several other general properties of hierarchical control systems that hold at all levels. The selection of the classification rules or the degree and type of detail that is ignored in a hierarchical constraint depends on the effectiveness of the particular function that is thereby established. In particular, when the constraints are too numerous or tight, or when they are too scarce or loose, function cannot be maintained. This is our second principle of Optimum Constraint.

Finally, the general property of biological organizations is that they appear to have an indefinite capability to evolve new functions and new hierarchical levels of control while

maintaining a relatively fixed set of elementary parts at each level. In other words, the variety of alternative descriptions and functions does not appear to be limited by the fixed set of structural elements that make up the constraints of the organization. We call this Statistical Closure.

A physical theory of the origin of hierarchical control levels would be a derivation of these principles from a combination of the existing fundamental laws, both dynamical and statistical. It would explain how complex collections of interacting elements spontaneously separate out persistent and coherent descriptions and functions under the constraints that relate them. The origin of life is the lowest level of this process where the genotypes (descriptions) and phenotypes (functions) are generated by the constraints of a genetic code. As yet such a physical theory does not exist.

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HIERARCHY THEORY

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of Complex Systems*

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